

Field Test Comparison of Wingtip Motor Aircraft Configuration

Kyle Russell, Sam Nelson, Frank Robles, Raisha Hussain, and Humza Khan

University at Buffalo, Buffalo, NY, 14261, United States of America

I. Executive Summary

This project investigates the effect of wingtip-mounted propellers on the overall power consumption and flight performance of an electric RC aircraft. While previous experiments verified the overall reduction of induced drag in a wingtip-mounted propeller configuration with a constant airspeed, the team could not find any research that focused on finding a correlation between constant throttle input and induced drag based on our proposed configurations in a flight test. In a racing environment, maximizing both airspeed and endurance is critical. As a result, a racing circuit with tight, steep-bank turns and loops will be used to measure the changes in airspeed, current, and power as the aircraft flies through the circuit in two different configurations (wingtip mounted motors vs. mid-wing mounted motors). When considering different aircraft designs, factors such as manufacturability, flight stability, budget, and pilot's experience required the construction of an aircraft that had a rectangular wing with washout, conventional tail, and tricycle landing gear.

II. Introduction

Members:

1. **Sam Nelson:** I am the project leader and pilot. My responsibilities include organization of tasks, executive decision making, and flying the plane. I also aid in manufacturing and design. I modeled much of the plane in CAD and facilitated technical discussions for the conceptual design of the aircraft.
2. **Kyle Russell:** My role was collaborative for the electronics of the aircraft, in which I learned how to set up INAV and how to properly install schematics for the electronic layout of our flight controls. I assisted in the process of pasting foam filler into the crevasse of the wing and then sanding it down to create a surface with no indents. I also aided in the manufacturing of the hotwire shaping of the fuselage.
3. **Raisha Hussain:** My role is to ensure all project stages are thoroughly documented, including the design changes, and manufacturing process. This includes delivering progress reports and creating detailed 2D drawings, ensuring all team efforts were documented with accuracy for future reference and reporting purposes. I actively contributed to the manufacturing of the wings and fuselage, ensuring the parts met design requirements and were ready for integration into the final product. I worked on the literature review, validation, and recommendation section.
4. **Frank Robles:** My main role is FPV observer and data collection during flight tests by recording all flight videos. I have also collaborated in the manufacturing process of the wings and stabilizers by importing 2D airfoil templates and generating CNC codes for cutting out all the pieces out from EPS foam. I have also set up pitot sensors for the flight test.
5. **Humza Khan:** One of my roles was to source funding and keep track of the team expenditures. Furthermore, I aided in the manufacturing of the main fuselage by cutting foam blocks with the hotwire

cutter. I was also responsible for designing parts like servo motor mounts needed to securely install servo motors for control surfaces of the aircraft. I am also going to be the third-person observer of the test flight.

Funding and Sourcing:

Our project was funded in part by the \$250 allowance for each group. The remaining cost, roughly \$200, was split among members on a volunteer basis. Note that many parts and tools did not need to be purchased and are excluded from the budget because they were already owned by our members or borrowed from UB AIAA's Design Build Fly team.

Although the team set up a GoFundMe page, and pursued funding through ELN, it was concluded that funds from such sources would take longer to be received or would be left at the chance of not securing adequate funding through the donation platform.

The items were split under different budget pools as shown in Tables 1 and 2.

Parts	Quantity	Cost/item	Total cost
Epoxy	1	\$28.99	\$28.99
2.8 oz Fiberglass	1	\$16.99	\$16.99
0.85 oz Fiberglass	1	\$11.99	\$11.99
Landing Gear	1	\$24.38	\$24.38
Airspeed sensor / Pitot tube	1	\$56.00	\$56.00
GPS Sensor	1	\$16.99	\$16.99
Antennas	1	\$22.99	\$22.99
Flight Controller	1	\$55.99	\$55.99
M3 screws + nuts	1	\$6.99	\$6.99
Control Arms	1	\$7.99	\$7.99
		Grand Total	\$249.30

Table 1: Budget list with the \$250 allowance

Parts	Quantity	Cost/item	Total cost
Batteries	2	\$47.95	\$95.90
Motors 550KV	2	\$30.00	\$60.00
2x12 APC Propellers (Pusher)	2	\$5.00	\$10.00
2x12 APC Propellers (Puller)	2	\$5.00	\$10.00
		Grand Total	\$175.90

Table 2: Budget list with the remaining cost

Importance of the project:

The topic of wingtip propulsion is of great interest to UB's Design Build Fly team, because wingtip motors may offer an advantage in the competition. The results of our flight tests should inform whether or not to implement wingtip propulsion in our competition aircraft, and how to best implement it.

Efficient ways to integrate electric propulsion with the airframe is also relevant to future sustainable aircraft projects, especially in the urban taxi and STOL markets. Our flight tests will help quantify the performance benefits of this configuration, as well as the practical considerations such as the likelihood of prop strikes and stall characteristics, which could be relevant not only to DBF but also to broader general aviation projects.

Decision-Making Process

We made decisions based on the project requirements, and came to solutions after having group discussions on which option is viable or offering an alternative option. Our decisions were informed by general airplane performance calculations or conventional rules of thumb for RC airplane design when applicable.

Much of our design was driven by budget and time constraints. For example, we decided to use a simple and cheap open-source flight controller setup that is reliable for data collection on UAVs. Research concluded that for our level of requirements, INAV was capable of getting basic accurate flight data with little setup and troubleshooting process.

III. Project Background

Requirements:

In order to evaluate how throttle input impacts on efficiency in a wingtip motor aircraft, accurate data is required for propulsion and airspeed, while keeping other variables such as weight and power the same. We require the aircraft to have a considerable low drag while obtaining a highly maneuverable configuration which can reach a minimum speed of 80 mph. This speed simulates real world AIAA DBF competition requirements where having the fastest lap time on a flying circuit will guarantee a higher score position. We have a goal of making the UAV achieve 100 mph top speed which we see as a major accomplishment. We require the aircraft to operate safely throughout the flight path that results in a safe landing and the retrieval of the data collected in flight. We also require a minimum power to maintain flight. This is about 100 watts/pound by conventional RC guidelines, however, given our speed target we will likely have a much higher power requirement.

Constraints:

Hermes must be built and tested under the timeline of the 451 Design Processes class, where we have approximately 13 weeks to design, build, and fly our prototype aircraft. As the project continues into the winter season, the overall weather condition will worsen due to the high probability of snow/rain, which incentivises us to test as early as possible. In addition, the project group has a limited allowance of \$250 to allocate for building materials and sensors. As a result, some parts were purchased out of pocket. Lastly, while our pilot is skilled enough to reliably fly a medium sized sport aircraft (E-flite Beechcraft D18 1.5m BNF Basic), there are some concerns with our pilot flying a larger custom built RC plane during the flight test.

Operating Conditions:

All of the flight tests will be conducted at the Wing and Rotor club of Western New York. This field features a grass runway, roughly 200 feet long, with a roughly 1000-foot long area for flying. Our flying conditions will become colder as winter approaches, which will negatively impact battery health and performance. As such, we must be careful to record ambient temperature at each flight and interpret data with temperature in mind. All flight tests will ideally be conducted in one day, swapping between two different configurations of flight mode for experimenting and data collection. The flight circuit path of our aircraft will follow along the path as shown below.

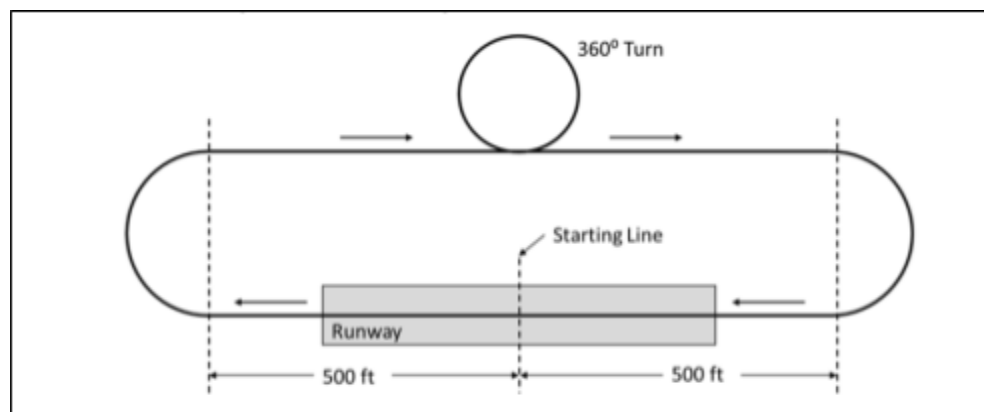


Figure 1: Aircraft Flight Circuit

Deliverables:

The deliverables for the project include a comprehensive set of outputs that ensure both the design and performance evaluation of the aircraft. These consist of a detailed 3D model and 2D technical drawings, the final physical aircraft, and the setup of the flight controller and ground GPS testing. Key metrics such as speed versus power consumption, the number of laps relative to battery life, including voltage, current, airspeed, RPM, ground speed, altitude, and plane orientation, will be analyzed. Additional deliverables include a bill of materials outlining all required components and materials, along with weekly progress reports that document the project's ongoing developments and milestones. As a minimum success criteria, the aircraft needs to land and survive multiple experimental flights in order to guarantee data for comparison.

IV. Technology and Literature Review

The following two literatures were used in this project:

“Verifying the Effect of Wingtip Propellers on Drag Through In-Flight Measurements”

The study by Pfeifle et al. (2021) in “ Verifying the Effect of Wingtip Propellers on Drag Through In-Flight Measurements”, highlights the aerodynamic benefits of wingtip-mounted motors. Their findings of an 8-15% reduction in drag with wingtip-mounted propellers are directly applicable to the design and analysis of this RC aircraft project. By adapting their methodologies, our project will contribute further insights into how motor placement impacts the performance of small-scale electric aircraft, particularly regarding energy efficiency and drag reduction.

Pitot tubes are essential tools for accurately measuring airspeed. In the study conducted by Pfeifle et al. (2021), a five-hole air data probe was used to measure airspeed and aerodynamics angles , enabling precise analysis of how the propulsion system interacts with the aircraft's aerodynamic forces. This is especially useful in comparing the efficiency of wingtip-mounted versus mid-wing-mounted motors. For our project, using a similar sensor would provide valuable real-time data on the aircraft's airspeed in different configurations. However, due to limited funding, we are unable to incorporate these high-precision instruments into our experimental setup. Instead, we will be installing a PX4 differential airspeed pitot tube directly in front of the fuselage, as this fits better within our budget while providing the performance we need.

In our project, we aim to explore how varying power settings affect the efficiency of wingtip-mounted motors compared to mid-wing-mounted motors. This new direction will allow us to investigate more nuanced aspects of propulsion efficiency, such as the relationship between throttle settings, drag, and power consumption during flight. This refined focus not only complements the existing data from studies like Pfeifle's but also expands our understanding of motor placement and energy efficiency in RC aircraft.

“Wingtip-Mounted Propellers Aerodynamic Analysis of Interaction Effects and Comparison with Conventional Layout”

The study by Sinnige et al. (2019) in "Wingtip-Mounted Propellers: Aerodynamic Analysis of Interaction Effects and Comparison with Conventional Layout." demonstrates how wingtip-mounted propellers can reduce drag, improve wing efficiency, and enhance overall aerodynamic performance, over the conventional inboard-mounted configurations.

One of the key findings in the literature is that wingtip-mounted propellers significantly reduce induced drag by attenuating the wingtip vortex through the propeller slipstream. A 15% drag reduction was observed under specific

lift and thrust conditions. This configuration enhances the wing's effective span efficiency, especially when the propeller rotation opposes the wingtip vortex (inboard-up rotation), leading to favorable aerodynamic interaction. The drag reduction observed with wingtip-mounted motors implies that less throttle input is required to maintain the desired flight speed. This can optimize power usage and improve overall efficiency, which is crucial for meeting the performance goal of reaching 100 mph with minimal energy expenditure. The ability to achieve high efficiency with lower throttle input is also beneficial in ensuring the aircraft remains within the power constraints, particularly the 250 watts/pound requirement to prevent stalling.

Another one of the key findings in this literature is that compared to conventional layouts—where propellers are mounted inboard along the wing—the wingtip-mounted configuration demonstrated superior aerodynamic performance, particularly in drag reduction and overall efficiency improvements. Tests conducted in a low-speed wind tunnel, supported by particle image velocimetry (PIV) data, confirmed these benefits, showing reduced swirl and improved flow characteristics in the propeller's wake. Early research by Snyder and Zumwalt (1960s) demonstrated that wingtip-mounted tractor propellers could enhance the lift-to-drag ratio by mitigating wingtip vortices. Sinnige et al.'s study builds on this foundation, providing detailed experimental and numerical evidence of how factors such as propeller rotation direction and thrust coefficient influence these aerodynamic benefits. On the other hand, conventional inboard-mounted propeller configurations show substantial aerodynamic interactions due to the propeller slipstream's influence on the wing. Previous studies referenced in this literature have shown how this slipstream can affect local lift and drag distributions, offering critical insights for aircraft design, especially as hybrid-electric propulsion technologies expand the range of propeller integration options.

Other design alternatives worth considering include distributed propulsion systems, which employ multiple smaller motors along the wing span to optimize airflow and efficiency. Additionally, pusher configurations—where the propellers are located aft of the wing—may introduce distinct aerodynamic interactions compared to the tractor setups used in this study. However, these alternatives pose challenges in terms of structural complexity, aeroelasticity, and control during single-engine failures.

Patents

In terms of patents, there are no substantial premade RC products we are using as our experimental aircraft. All structural components and airframe for our aircraft are custom made from EPS foam and fiberglass composite with epoxy. Some patents that may be similar to our project would be the WO-2021138470-A1 which is an aircraft with wingtip positioned propellers but the intended design was for a VTOL configuration where the wingtip nacelle is a tiltrotor. Overall, the resulting patents don't have an overall impact on our design methodologies.

V. Design Methodology 1

All of the team members are active members in UB AIAA - Student Chapter and have extensive experience designing and building experimental RC aircrafts based on DBF competition rules and guidelines. As the design challenges from DBF changed on a yearly basis, two years ago, an innovative approach was adopted that employed wingtip mounted motors for the team's aircraft, Hammerhead, after finding a research paper from DUT which indicated that wingtips mounted propeller resulted in significant reduction in induced drag of the aircraft. Even though Hammerhead was a successful prototype that fulfilled the DBF goals for that year, the DBF team did not collect flight data during the test flights. The main goal of the project was determined to remedy this shortfall, and help quantify the benefits of the wingtip motor configuration in a DBF setting.

Since Hammerhead was a unique, unconventional design (comprising of a V-tail, Tapered wings, and a blended-body fuselage), we decided that Hermes needs to be as conventional as possible so that consistent results on flight performance can be obtained and standardized for the application of the concept on other RC aircrafts.

The team set up weekly meetings where all members brainstormed on ideas after the main goal(s) of the project were set. The meetings consisted of members making rough drawings of what the prototype of the aircraft is going to look like. Concepts were evaluated through a process by which the members logically critiqued the potential weaknesses of that design based on factors like manufacturability, budget, DBF historical data, and pilot's comfort

with the presented aircraft style. A variety of concepts were considered for both sizing and configuration, with our current design being decided as best meeting the aforementioned factors. Ultimately, specific configuration decisions were driven by specific requirements. For example, the use of a pitot tube on the nose of the aircraft makes a tricycle landing gear more desirable, since it reduces the chance of the pitot tube being damaged on landing.

V-tails and T-tails were both considered, but ultimately a conventional tail was selected. Originally, it was believed that a V-tail or T-tail would keep the tail out of the propwash and thus provide a more fair comparison between the motor configurations, as the inboard-mounted propeller slipstream would interfere with a conventional tail. However, this was later decided to be a benefit, as it would help more realistically determine potential practical effects of each configuration.



Figure 2: Wing Assembly

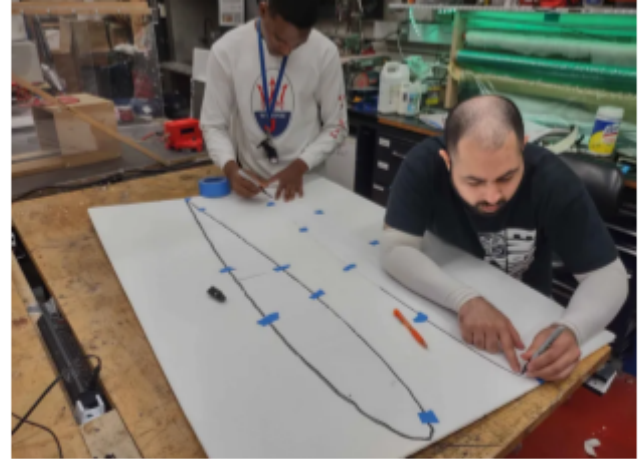


Figure 3: Cutting foam-board templates for hotwire cutting the fuselage

VI. Design Methodology 2

Engineering Process

Once our configuration was selected, we selected a wing chord which would allow the plane to operate at Reynolds number similar to DBF competition planes ($Re = 500,000 - 1,000,000$). This defines our MAC and cruise speed. A rough wing loading was then selected, as per our design requirements which must reflect a typical DBF aircraft. With weight and wing area defined, we then sized the tail according to a 0.5 tail volume coefficient, which is historically a good balance of stability and maneuverability.

The design was then modeled in CAD using Fusion360. This is mandatory for 3D printing. Additionally, paper templates for the fuselage were created by tracing the shape off of a laptop's screen (scaled to 100%). CAD modeling also allows for accurate weight and CG estimates, which is critical for good aircraft design.

The experimental setup will gather data from a typical hobbyist-grade UAV flight controller fitted with an airspeed sensor, GPS module, and voltage/current sensors, and log data onto an on-board a microSD card, which is recorded in .dat format.. The electronic speed controllers are used to gather RPM data from the motors, which can be used to find our thrust at any given RPM value from the manufacturers data. This setup should measure almost all variables relevant to airplane performance as a function of time since takeoff. This allows for accurate analysis to be conducted and conclusions to be drawn from the data.

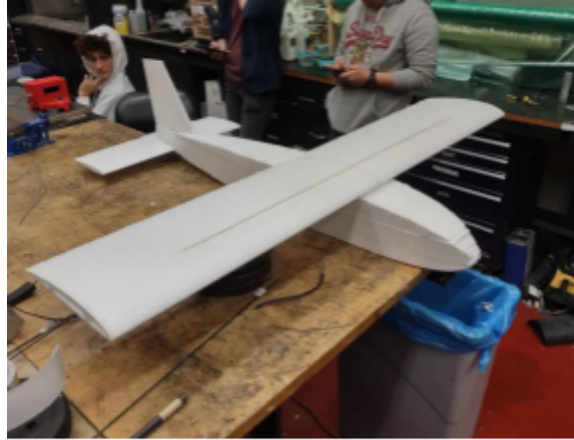


Figure 4: Hermes Full Loose Assembly

Assembly and Fabrication Methods

The Fabrication process started with hotwire cutting our airfoils and fuselage out of rectangular pieces of foams. We then apply 'foam filler' to fill the gaps. The foam parts are then sanded, and plastic hardpoints are inserted. Then, we perform a fiberglass composite layup over the foam, which is then filled and sanded. We repeated the process multiple times to get the surfaces as smooth as possible.

The nose gear is an off-the-shelf product of spring-steel wire construction. A fiberglass landing gear arch is used for the main gear, which is screwed into wooden hardpoints underneath the fuselage. 3" foam wheels are fitted, as this is typically the minimum wheel size which works well on grass runways.

Servos and electronics will be installed using 3D-printed hardpoints and brackets, printed from PLA. Motor mounts will be printed from ABS for its superior thermal stability, since our motors may exceed 60°C, the glass transition temperature for PLA.

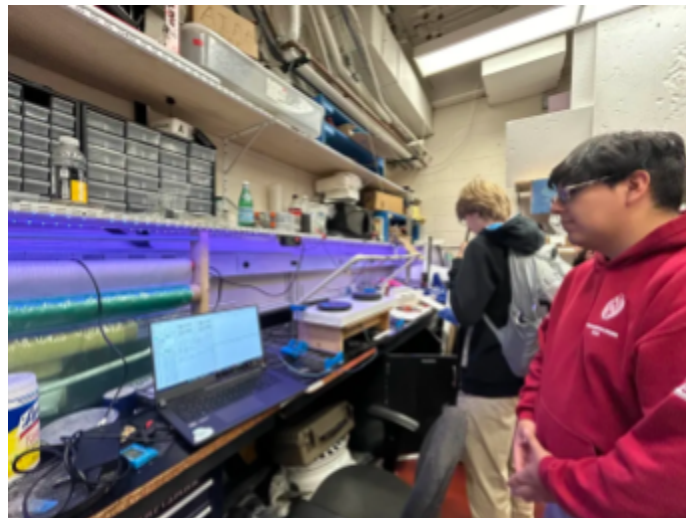


Figure 5: Hotwire Cutting Wing Sections

VII. 3D assembly Drawing

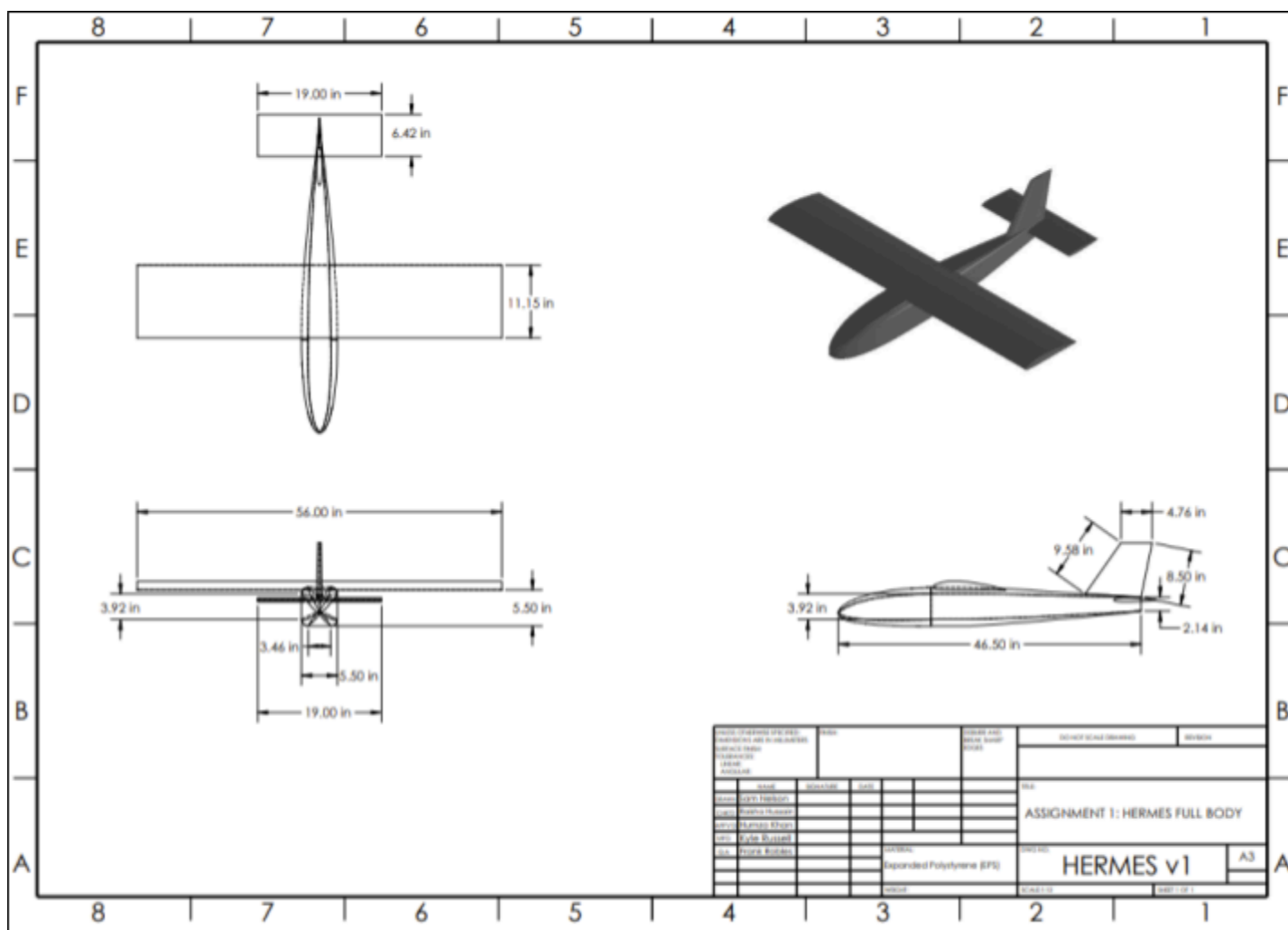
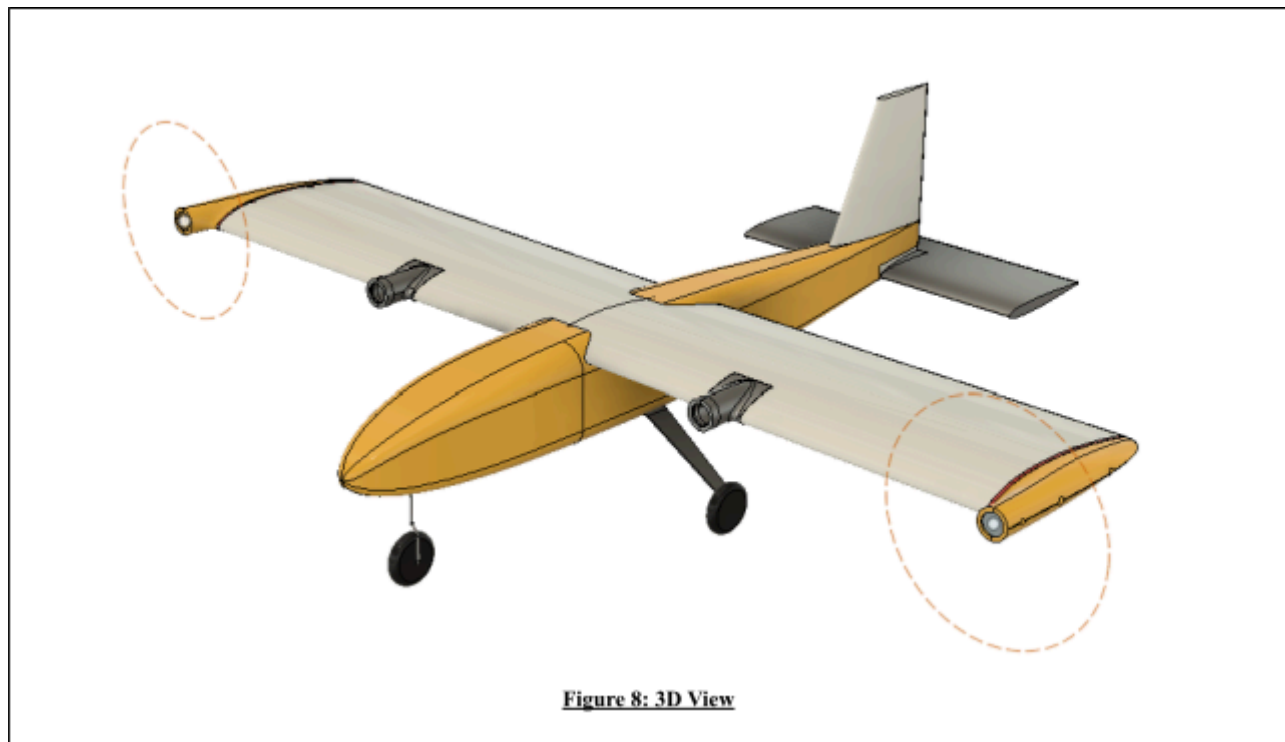
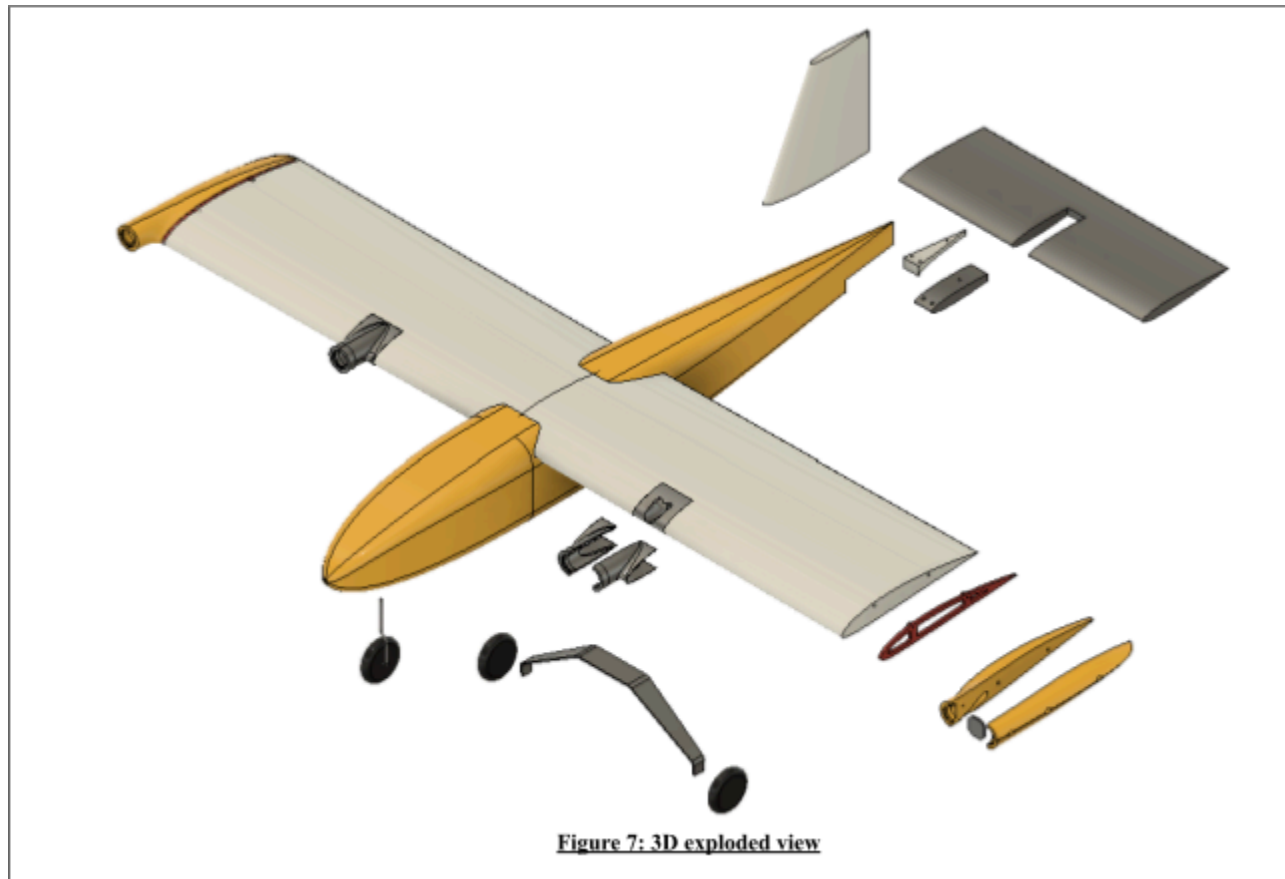


Figure 6: Engineering Drawing



VIII. Engineering Analysis

Stability:

Stability was calculated by use of tail volume and static margin. Our tail was sized to a 0.5 volume coefficient, and a static margin value of 10% was selected. Using a free online calculator, our CG position was calculated. While this does not constitute a full stability analysis, our experience indicates it is sufficient provided a negative angle of incidence is present between the wing and the tail.

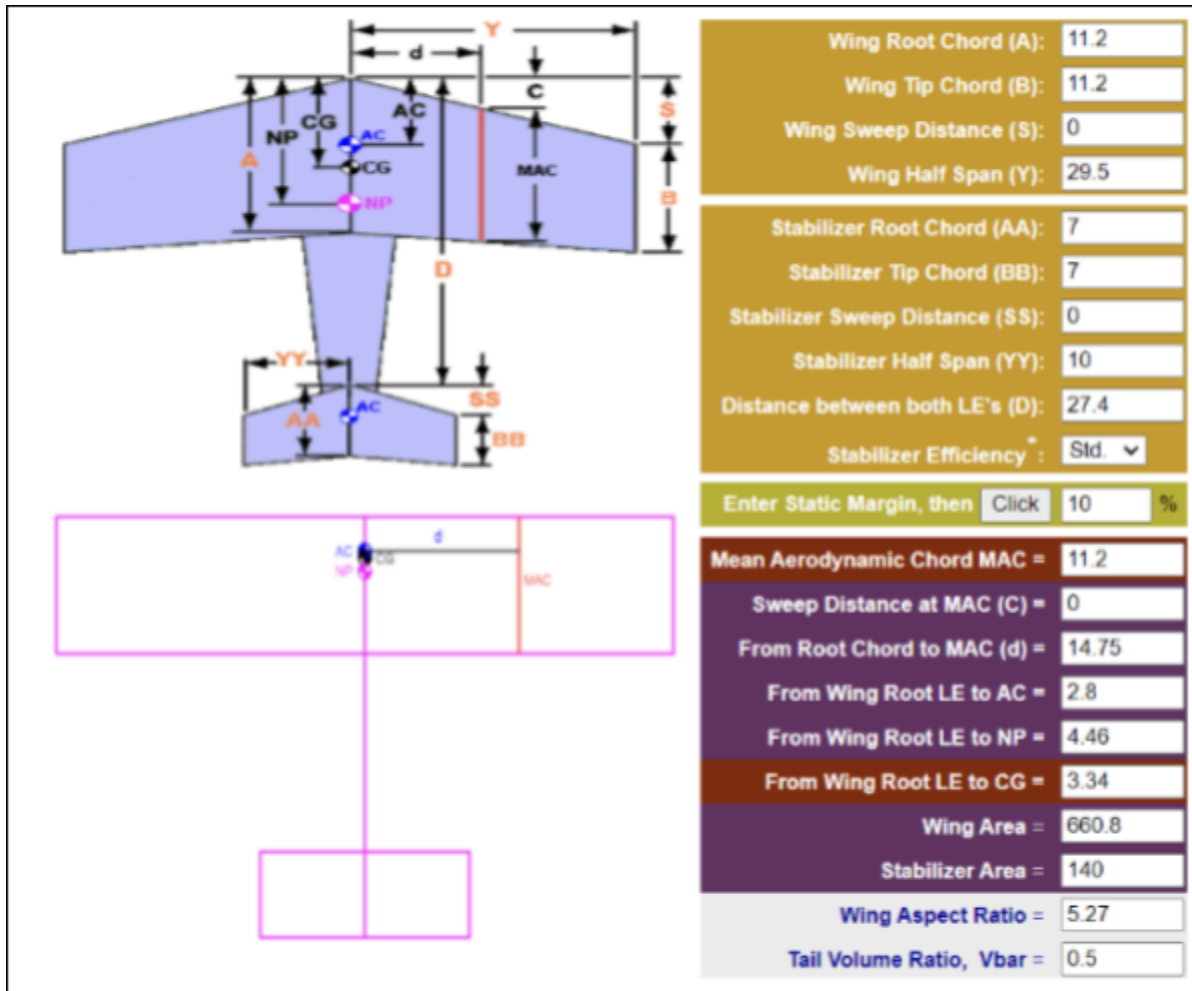


Figure 9: Stability Analysis

Stall speed/Wing loading:

Our stall speed was calculated using an Excel spreadsheet utilized by UB DBF according to the analytical stall speed equation: $V = \sqrt{2w / \rho S C_{L, \max}}$. A loose threshold of 25 mph was set as a target stall speed for easy handling. Takeoff distance was also calculated, just for fun. This calculation informed our wing sizing, through an iterative approach. This stall speed assumes an infinite wing, so our maximum lift coefficient should be lower, and consequently, our stall speed will be higher than predicted here.

Variable	Value	Unit
Static Thrust	10	lbs
Weight	7	lbs
rho	0.00194	slugs/ft ³
Cl	1.4	-
Wing area	4.58	ft ²
Takeoff Distance	17.61	ft
V _{stall}	33.5	ft/s
V _{stall}	22.8	mph

Table 3: Stall Speed Calculation

eCalc:

Our propulsion system was chosen using eCalc, an online electric drive calculator. By entering parameters for the motor, propeller, and battery, our propulsion performance can be predicted. Our selected motor and propeller combination results in a peak power of 1100 watts, at roughly 56 amps and 20 volts. This is just barely within the limits of our motor and ESC ratings.

eCalc also predicts speed, based on a simplified drag model using wing area and drag coefficient. Our predicted maximum speed is 95 mph, as per Figure X. However, eCalc historically underestimates speed by a significant margin, because it assumes a fixed drag coefficient of 0.3. This suggests that our maximum speed may be in excess of 100mph, far exceeding our target DBF speed estimate of 80mph.

Propeller	Throttle	Current (DC)	Voltage (DC)	el. Power
rpm	%	A	V	W
2400	21	0.9	22.2	20.8
3000	26	1.7	22.1	37.6
3600	32	2.8	22.1	62.4
4200	37	4.4	22.0	97.0
4800	43	6.6	22.0	143.3
5400	50	9.4	21.9	203.3
6000	56	12.9	21.7	278.9
6600	63	17.4	21.6	372.3
7200	70	22.9	21.4	485.6
7800	78	29.7	21.2	621.1
8400	86	37.8	20.9	781.0
9000	94	47.7	20.5	967.7
9342	100	54.9	20.3	1100.4

Table 4: eCalc (RPM & Throttle)

Thrust		Spec. Thrust		Pitch Speed		Speed (level)	
g	oz	g/W	oz/W	km/h	mph	km/h	mph
331	11.7	15.6	0.55	44	27	-	-
518	18.3	13.4	0.47	55	34	35	22
746	26.3	11.6	0.41	66	41	59	37
1015	35.8	10.2	0.36	77	48	69	43
1326	46.8	9.0	0.32	88	55	79	49
1678	59.2	8.0	0.28	99	61	89	55
2071	73.1	7.2	0.25	110	68	98	61
2506	88.4	6.5	0.23	121	75	108	67
2982	105.2	6.0	0.21	132	82	118	73
3500	123.5	5.5	0.19	143	89	128	79
4059	143.2	5.0	0.18	154	95	138	86
4660	164.4	4.7	0.16	165	102	148	92
5004	176.5	4.4	0.16	171	106	153	95

Table 5: eCalc (Thrust)

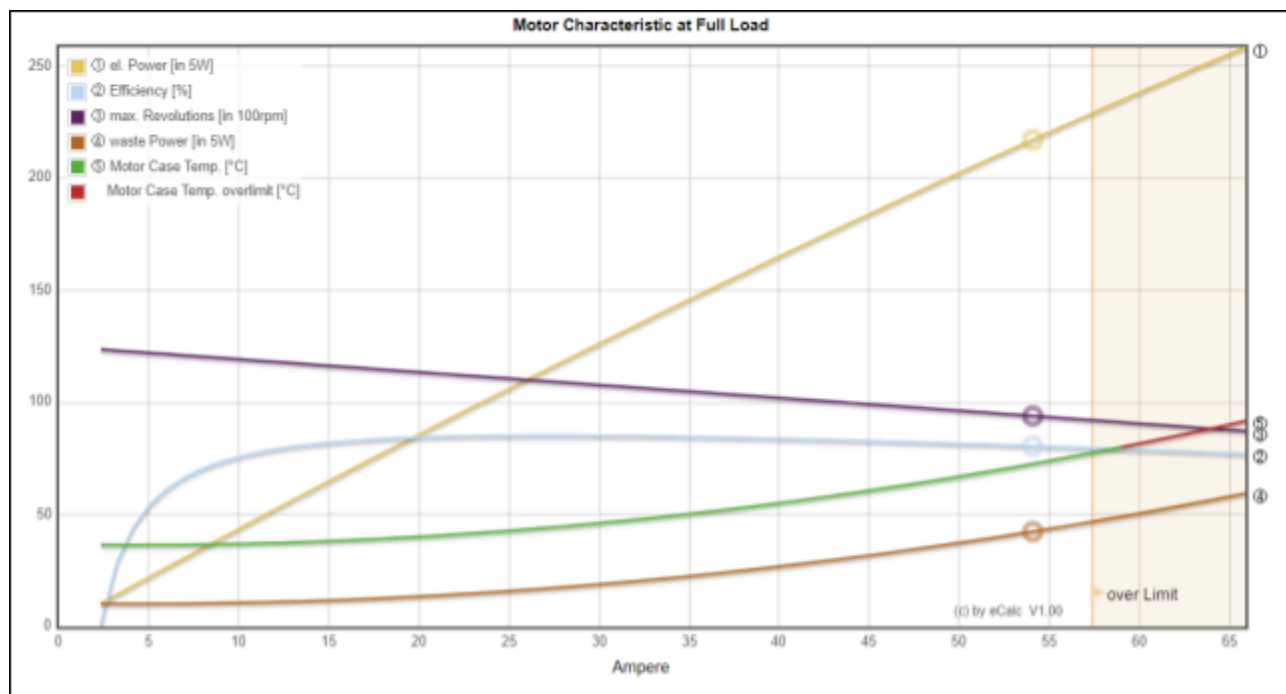


Figure 10: eCalc (Shows that the selected motor is always under the critical Ampere value)

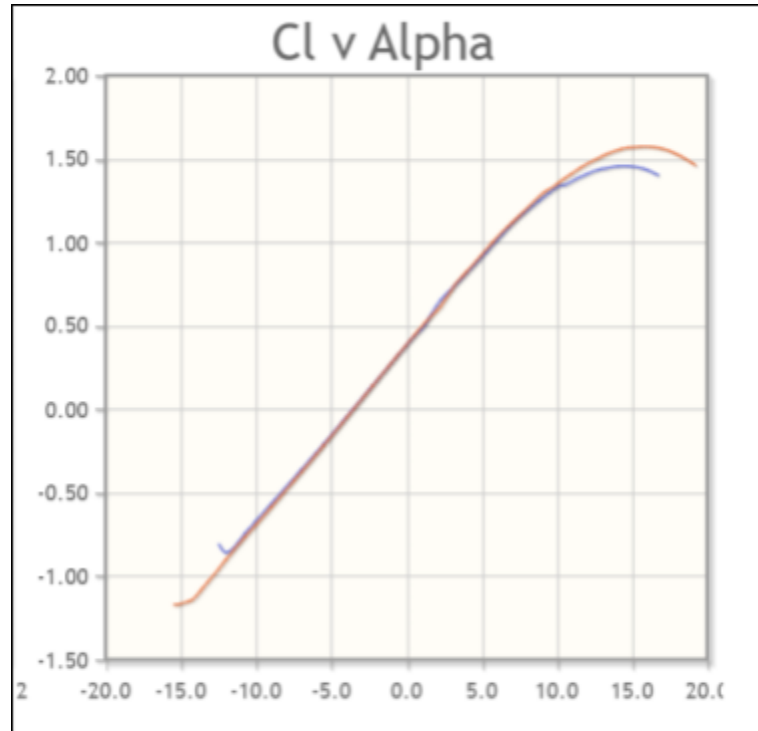


Figure 11: Lift curve slope for Clark Y airfoil

IX. Validation

The validation phase involved testing the RC aircraft to assess its ability to meet the project's performance goals and evaluate the effects of motor configurations (inboard vs. outboard) on flight performance. These tests aimed to determine whether wingtip-mounted motors, theorized to reduce induced drag by mitigating wingtip vortices, would provide superior flight performance. Testing was conducted at Reservoir State Park in Niagara Falls under, with five test flights completed. Before each flight, the aircraft was assembled into the required configuration using hand tools, and pre-flight checks of the propulsion and control system were performed.

A. Testing Results

The key metrics recorded during the validation tests include airspeed, throttle input, and power consumption. These parameters were logged using an onboard flight controller, GPS module, and pitot tube. A comparison of the inboard (conventional) and outboard (wingtip-mounted) motor configurations yielded the following results:

- 1) **Inboard Configuration (Conventional):**
Average airspeed: 72 mph over the test circuit
- 2) **Outboard Configuration (Wingtip):**
Average airspeed: 71 mph over the same test circuit

Contrary to expectations, the wingtip-mounted configuration did not outperform the conventional inboard setup. While both configurations demonstrated favorable flight characteristics, the inboard motors achieved a slightly higher average airspeed. It should also be noted that the outboard configuration had a lower roll rate, qualitatively reported by the pilot. This is probably the result of an increase in moment of inertia around the roll axis. Fortunately, this did not seriously impact handling, even for a highly maneuverable aircraft like Hermes.



Figure 12: HERMES FPV Video Snapshot

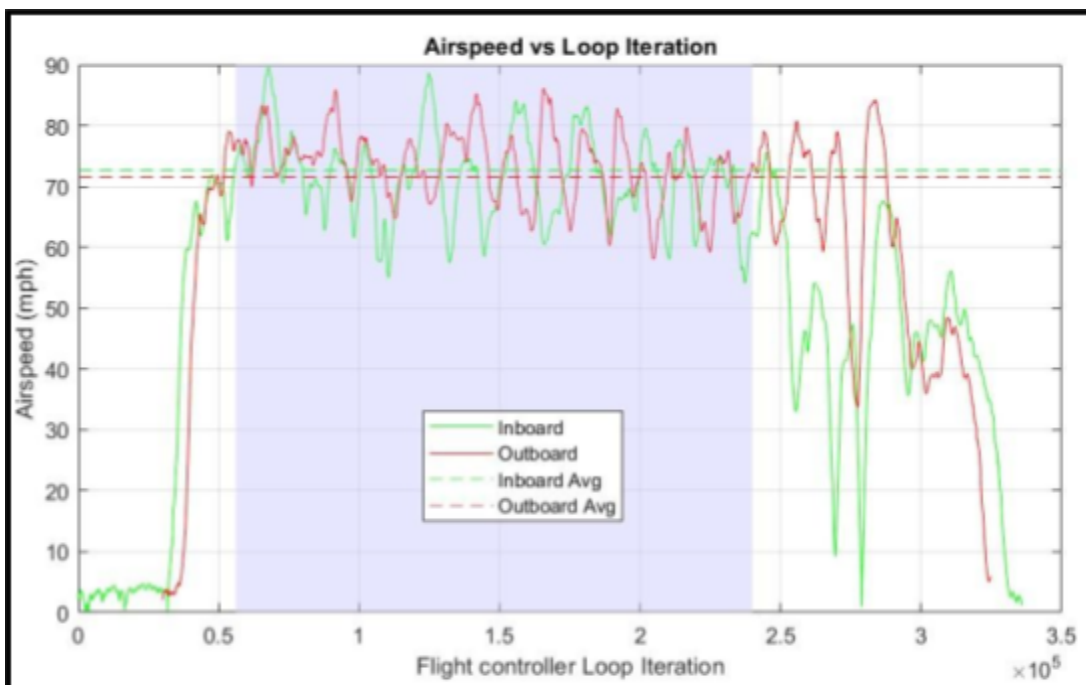


Figure 13: Flight Performance Data Plot

X. Performance Analysis

The project's hypothesis was based on established aerodynamic theory that wingtip-mounted propellers could reduce induced drag by interacting with and mitigating wingtip vortices. Wingtip vortices, generated by pressure differences between the upper and lower surfaces of a wing, contribute significantly to induced drag. By placing propellers at the wingtips, the propeller slipstream can disrupt these vortices, reducing drag and increasing overall aerodynamic efficiency.

Despite this theoretical advantage, the wingtip-mounted configuration in this experiment did not perform as anticipated. The marginally lower airspeed observed in the outboard configuration raises questions about the practical implementation of this design. Two factors are most likely the cause of the discrepancy:

- **Manual Piloting:** Due to the nature of our flight test procedure, the flight path taken between each flight was not guaranteed to include the same load factors during turns or even the same number of total maneuvers. This could affect airspeed, because if one flight had more dynamic maneuvers, it would appear to have a lower average airspeed.
- **Limited Data Collection:** Our sample size was relatively small due to the scope of this project. Ideally, we would have hours of flight data rather than minutes. This magnifies any inconsistencies and leads to less accurate results.

XI. Engineering Standard

There are minimal engineering standards in the UAV industry as compared to the automotive or general aviation industry. This is likely due to the low risk and low cost of UAVs, as well as the time required for regulation to catch up with technology. However, the DBF competition does include a set of rules that aircraft must comply with to be legal. We will comply with most of the electronics safety and requirements from the DBF guidelines, for the sake of keeping our experimental results relevant to the DBF competition. Some of the major standards include:

- The airplane must be propeller driven and electric powered with an unmodified off-the-shelf electric motor. May use multiple motors and/or propellers. May be direct drive or with gear or belt reduction.
- There can be a maximum of one battery pack connected to a propulsion system. A propulsion system consists of one battery, one externally accessible arming fuse, one or more electronic speed controllers (ESC), and one or more motors.
- Propulsion power total stored energy cannot exceed 100 Watt-hours.
- Teams may choose between NiCad/NiMH OR Lithium based batteries.

Some of these rules impacted our design, such as limiting us to using sub-100Wh LiPo batteries.

Above all, FAA regulation must be complied as well as HERMES is identified as a small Unmanned Aircraft System. As a small UAS aircraft, the weight must not surpass 55 lbs, and cannot fly above the 400 ft. altitude mark. Any flights conducted at night require anti-collision lights, so we plan to fly exclusively during the daytime. Additionally, sUAS systems are required to be equipped with remote identification technology (RID), unless operated at a safe-fly-zone (called a FRIA) maintained by a community organization. This is a non-issue, because our local flying field is a registered FRIA.

XII. Recommendation

Design Optimization:

Refine the wingtip-mounted propeller design using advanced Computational Fluid Dynamics (CFD) tools. Focus on optimizing critical parameters such as propeller rotation direction, placement, and size to reduce drag and improve performance. Additionally, explore alternative wing geometries and nacelle configurations to enhance the interaction between the propeller slipstream and wingtip vortices, addressing inefficiencies observed during testing.

Flight Testing Enhancements:

Incorporate autonomous flight systems to standardize flight paths and eliminate variability caused by manual piloting. Waypoint navigation would ensure consistent and repeatable maneuvers, leading to more reliable comparisons. Conduct additional test flights under varied environmental conditions to expand the dataset and improve statistical confidence in the results, ensuring a more comprehensive understanding of motor performance in racing environments.

Instrumentation and Data Collection:

Upgrade the data logging system to ensure reliability and prevent data loss during testing. Introduce additional sensors, such as thrust load cells and advanced air data systems, to capture real-time thrust and aerodynamic forces. This expanded data collection would provide deeper insights into the relationship between motor placement, drag reduction, and overall performance.

Environmental Considerations:

Perform future tests under controlled conditions to minimize the effects of temperature, wind, and other environmental variables on motor and battery performance. Incorporate seasonal and location-specific weather patterns into test planning to ensure consistent conditions across experiments.

Broader Applications and Experimental Configurations:

Explore the scalability of wingtip-mounted propulsion for larger UAVs and general aviation applications, where drag reduction benefits may be more pronounced. Additionally, investigate alternative configurations, such as distributed propulsion systems and hybrid-electric designs, to evaluate their potential for improving efficiency and reducing drag in high-performance scenarios.

XIII. Conclusion

The flight test results have shown that the inboard configuration has achieved a higher average airspeed than the outboard configuration by one mph. The performance difference is negligibly small that a strong correlation between wingtip prop performance and aerodynamic efficiency cannot be made and additional tests must be performed for the aircraft to find a stronger difference in performance. Some factors that may have lead to the errors in data results could have been:

Rough Weather Conditions- As the project was developed more into the winter season, the overall weather conditions have worsened since September. All flight tests had to be performed on windy days as the best weather available in the last two weeks available for us to produce data. Wind gusts reached from 10-15 mph on test flight day which could have impacted the aircraft's stability to keep a stable flight path as seen from the data.

Low Overall Payload - While the low payload of HERMES allow the aircraft to be more maneuverable at high speeds, the low weight may have impacted the total induced drag that may have been created overall as not many lift is required to achieve takeoff especially when the aircraft has a power-to-weight ratio of four. Additional tests where more payload mass added to the aircraft can be made to verify the influence of mass in creating more induced drag and seeing more effect of the wingtip motors acting onto the vortices.

Overall, the 5 consecutive successful landings made by the HERMES aircraft is a success to the team as the aircraft is capable of handling more experiments to provide further tests to the DBF competition such as handling more payload and even installing additional attachment mechanisms to the fuselage to house a small autonomous glider to test the attachment mechanism and gliders' flight capabilities.

XIV. Acknowledgements

This research is dedicated to the organizers and volunteers working with AIAA's Design Build Fly competition. Design Build Fly has inspired all of us to pursue the art of aircraft design and construction in a greater depth than we knew we were capable of. This research is a testament to that drive to create.

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